Student Marks Analyzer using NumPy

A beginner-friendly Python project that generates marks for 50 students across 5 subjects and analyzes student performance using NumPy arrays, slicing, Boolean masking, and array operations.

Project Overview

This project creates a `50 × 5` marks array for the following subjects:

• Python
• NumPy
• Pandas
• Matplotlib
• Machine Learning

The marks are generated randomly, so the results may change each time the notebook is run.

Features

• Generates random marks for 50 students in 5 subjects.
• Displays array information: shape, number of dimensions, total elements, and data type.
• Uses NumPy slicing to display:
- The first 10 students
- Marks for the last subject
- The first 3 subjects for the first 10 students
- The last 5 students
• Uses Boolean masking to identify marks below 40 and above 90.
• Counts marks in grade bands:
- Below 40
- 40--59
- 60--79
- 80 and above
• Finds the highest mark and topper student number for each subject.
• Calculates the number of students passing each subject and each subject's pass percentage, using 40 as the pass threshold.
• Prints an overall summary, including average, highest, and lowest marks.

Technologies Used

• Python

• NumPy

• Google Colab / Jupyter Notebook

Getting Started

Option 1: Run in Google Colab

1. Open [Google Colab](https://colab.research.google.com/).
2. Upload `Student_Marks_Analyzer.ipynb`.
3. Run the notebook cell.

Option 2: Run locally

1. Install Python 3.

2. Install NumPy:

```
pip install numpy
```

3. Open the notebook in Jupyter Notebook or JupyterLab.

4. Run the notebook cell.

How It Works

1. NumPy creates an array with 50 rows and 5 columns.
2. Array attributes such as `shape`, `ndim`, `size`, and `dtype` describe the data.
3. Slicing selects particular students and subjects.
4. Boolean masks filter marks based on conditions.
5. `np.argmax()` and `np.max()` find the topper index and highest mark for each subject.
6. NumPy sums and percentages are used to calculate pass statistics and grade-band counts.

Important Notes

• Marks are generated using `np.random.randint(35, 100, (50, 5))`, so values range from 35 to 99.

• The project uses 40 as the passing threshold.

• Student numbers are displayed from 1 to 50, although NumPy array indices start at 0.

• The current notebook uses randomly generated sample data, not real student records.

Project Structure

```
Student-Marks-Analyzer/
■■■ Student_Marks_Analyzer.ipynb
■■■ README.md
```

Learning Outcomes

By completing this project, you can practice:

• Creating and inspecting NumPy arrays

• Indexing and slicing 2D arrays

• Boolean masking and conditional filtering

• Using aggregation functions such as `np.sum`, `np.mean`, `np.max`, and `np.argmax`

• Calculating pass percentages from array data

Future Improvements

• Use a fixed random seed so results remain reproducible.

• Add student names and roll numbers.

• Visualize subject-wise averages and pass percentages using Matplotlib.

• Export the analysis report to CSV.

Author

Add your name here.

If you found this project useful, consider giving the repository a star on GitHub.